Drift-Sense: Predicting IMU Dead-Reckoning Drift from Window-Level Motion Features

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Mid-semester report

1 Dataset and Exploratory Analysis

1.1 EuRoC MAV

Phase 1 uses two Machine Hall sequences from the EuRoC MAV dataset [1]: MH_01_easy and MH_02_easy. Each flight is a few minutes of a real hexacopter with a stereo camera and an ADIS16448 IMU. We use only `imu0/data.csv` (gyroscope in rad/s, accelerometer specific force in m/s^2, nominally 200 Hz) and `state_groundtruth_estimate0/data.csv` (position, Hamilton quaternion, velocity, and IMU biases). Camera images are reserved for Phase 2. MH_03_medium is not used in this phase.

1.2 Data quality

Table 1 summarises the two streams after loading. There are no missing values. Both IMU and the provided ground-truth estimate are sampled at 200 Hz with a 5 ms median step and negligible jitter. Nearest-timestamp alignment drops 1.17% and 1.32% of IMU rows (MH_01, MH_02), almost entirely at sequence edges where ground-truth coverage starts later than the IMU. We treat those drops as coverage gaps, not sensor faults.

Table 1: Sequence-level quality checks.

Sequence	IMU rows	Duration (s)	Rate (Hz)	Missing	Align drop
MH_01_easy	36820	184.1	200	0	1.17%
MH_02_easy	30400	152.0	200	0	1.32%

Raw gyro over the first 10 s of MH_01 is small-amplitude chatter ($\sim$0.2 rad/s), i.e. noise plus slow attitude change rather than aggressive spins. Accelerometer magnitude stays near g; the vehicle is pitched, so gravity appears on a_x and a_z together, not on a_z alone.

1.3 The two easy flights

The flights occupy a similar machine-hall volume. Mean ground-truth speed is 0.44 m/s on MH_01 and 0.49 m/s on MH_02 (peaks about 2.0 m/s and 1.2 m/s). Path length is $\sim$81 m and $\sim$74 m. Mean gyro magnitude is $\sim$0.21 rad/s on both. They are close enough that leave-one-sequence-out is a fair, if small, generalisation test, and different enough (duration, path) that holding one out is still meaningful.

1.4 Drift label

The headline target is the 2 s raw-IMU position error $y =$ `err_mag_m`. Across 720 and 592 overlapping windows the means are 1.215 and 1.203 m with standard deviations 0.058 and 0.079 m.

The histograms are tight and left-skewed (skew -1.7, -2.1; max 1.34 m). We therefore **do not** log-transform the primary target.

The same windows after subtracting the dataset's ground-truth IMU biases (`err_mag_m_bc`) have means 0.063 and 0.054 m. This supports the claim that 2 s raw drift is bias-dominated; the bias-corrected series is an oracle comparison, not the leaderboard target. Signed axis errors $(\Delta x, \Delta y, \Delta z)$ vary more than the Euclidean magnitude, so Phase 1 models should also be trained on the 3-axis target.

Mean error versus window length (0.25–2.0 s) grows superlinearly for the raw integrator ($\sim$6 mm at 0.25 s to $\sim$1.2 m at 2 s) and remains much smaller after bias correction. The primary window was changed from 1.0 s to 2.0 s (0.25 s stride) because the longer label is far more predictable (Section 4.5).

1.5 Train/test protocol

Windows overlap by 87.5%. A random split of rows would place nearly identical IMU slices in both train and test and inflate R^2. With two sequences we still use leave-one-sequence-out, recorded in `data/processed/splits.json`: train on MH_02 and test on MH_01, then swap. The leakage test `tests/test_no_leakage.py` asserts that no sequence appears on both sides of any fold. The primary target in that file is `err_mag_m`.

2 Method

2.1 Drift Label Construction

The EuRoC dataset does not provide a pre-computed odometry-drift signal; it supplies raw IMU measurements and precise ground-truth pose from a motion-capture system. We construct the regression target — the position error that a Visual-Inertial Odometry system would accumulate over a short time window — directly from these two streams.

2.1.1 Temporal Alignment

IMU measurements (200 Hz) and ground-truth pose estimates are recorded asynchronously. For every IMU sample we attach the temporally nearest ground-truth sample, discarding IMU samples whose nearest match exceeds a 20 ms tolerance. Across the two sequences used in Phase 1 (MH_01_easy and MH_02_easy), this discarded between 1.2% and 1.3% of IMU samples, concentrated at sequence boundaries where ground-truth coverage is sparser than IMU coverage.

2.1.2 Windowing

The aligned stream is divided into overlapping windows of $T_w = 2.0$ s (400 IMU samples at 200 Hz), with a stride of $T_s = 0.25$ s between consecutive window starts. This yielded 720 and 592 windows for MH_01_easy and MH_02_easy respectively. An earlier version used $T_w = 1$ s; Section 4.5 shows why the longer window was adopted.

2.1.3 IMU Strapdown Integration

For each window, we compute a model-free position estimate by integrating the raw IMU measurements forward using standard strapdown inertial mechanization. Starting from the true position p_0, velocity v_0, and orientation quaternion q_0 at the window's first timestamp (taken

directly from ground truth), we integrate at each IMU sample i with timestep Δt_i:

$$a_{\text{world},i} = R(q_i)\, a_{\text{body},i} + g \tag{1}$$

$$v_{i+1} = v_i + a_{\text{world},i}\, \Delta t_i \tag{2}$$

$$p_{i+1} = p_i + v_i \Delta t_i + \tfrac{1}{2} a_{\text{world},i} \Delta t_i^2 \tag{3}$$

$$q_{i+1} = q_i \otimes \Delta q(\omega_i, \Delta t_i) \tag{4}$$

where $R(q_i)$ is the rotation matrix corresponding to the current orientation estimate, g is the world-frame gravity vector, and $\Delta q(\omega_i, \Delta t_i)$ is the incremental rotation from the gyroscope reading ω_i.

Two sign/frame conventions — the sign of g and whether $R(q_i)$ maps body-to-world or world-to-body — are not fixed by the dataset documentation alone and were determined empirically: we evaluated all four combinations on a short (0.25 s) probe window against ground truth and selected the combination minimizing position error. This yielded $g = (0, 0, -9.81)\,\text{m/s}^2$ under a body-to-world rotation convention, with a probe-window error of 0.017 m — one to two orders of magnitude lower than the three alternative conventions (0.27–0.63 m), which we take as strong evidence the correct convention was identified rather than an arbitrary best-of-four selection.

2.1.4 Label Definition

The drift label for a window is the Euclidean distance between the integrated position estimate $\hat{p}_{\text{end}}$ at the window's final timestamp and the true position p_{end} from ground truth:

$$y_{\text{err}} = \|\hat{p}_{\text{end}} - p_{\text{end}}\|_2 \tag{5}$$

The signed per-axis error $(\Delta x, \Delta y, \Delta z)$ is retained alongside the scalar magnitude for use in the Phase 2 trajectory correction step and as additional Phase 1 regression targets. Across the two sequences, mean 2-second-window *raw*-IMU drift was 1.215 m (MH_01) and 1.203 m (MH_02), with maximum single-window error of 1.34 m (MH_01_easy). After oracle bias subtraction the same 2 s windows have mean residual 0.063 m (MH_01) and 0.054 m (MH_02), so roughly 95% of the raw drift is integrated IMU bias.

The same windows are also integrated after subtracting the dataset's ground-truth IMU bias estimates. The resulting columns (`err_mag_m_bc`, `err_dx/dy/dz_bc`) are an oracle comparison, not a replacement of the approved raw-IMU target: a deployed estimator is not handed those biases, and all features remain raw IMU. Whether the residual is more motion-dependent is measured in the EDA, not assumed.

2.1.5 Validation

We verified the label construction is physically sound by confirming that mean drift error increases with window length (0.25 s to 2.0 s), consistent with the expected accumulation behavior of open-loop inertial integration. This check, along with the convention-selection probe error, is automated in `tests/test_label_sanity.py` and passed on all sequences prior to any modeling work.

2.2 Feature Engineering

Each window is a 400×6 matrix of raw IMU samples (three gyroscope and three accelerometer axes over two seconds). Classic regressors cannot consume this raw tensor directly, so we summarise every window as a single fixed-length vector of 104 hand-crafted features. The guiding hypothesis is that open-loop inertial integration accumulates position error fastest in windows that are *energetic*, *erratic*, or *abrupt*; the feature families below are designed to quantify exactly those properties from the signal alone, using only raw IMU (no ground-truth information leaks in).

2.2.1 Time-domain features (60)

For each of the six channels we compute ten statistics: mean, standard deviation, variance, RMS, minimum, maximum, peak-to-peak range, skewness, excess kurtosis, and zero-crossing rate. The first-moment and energy terms (mean, RMS, variance) measure how much the drone moved and how large a bias the integrator must contend with; the shape terms (skewness, kurtosis) flag asymmetric or spike-laden windows; and the zero-crossing rate, computed about each channel's own mean so the accelerometer gravity offset does not dominate, is a cheap proxy for oscillation frequency. This family contains the single strongest linear correlates of the label, the accelerometer x-axis mean and RMS ($|r| \approx 0.67$), followed by the z-axis mean and RMS ($|r| \approx 0.54$), and supplies the top-ranked features under a RandomForest importance analysis (`ax_mean`, `ax_rms`). The accelerometer means mostly encode the window's tilt against gravity, which sets how the IMU bias projects into the world frame (Section 3.2).

2.2.2 Frequency-domain features (18)

Time-domain statistics describe *how much* a window moves but not *how* it moves. For each channel we remove the per-window mean, take a real FFT, and extract the dominant frequency, the total spectral energy, and the normalised spectral entropy of the power spectrum. Mean removal is essential: without it the accelerometer's large gravity/DC term would be read as a spurious 0 Hz peak. Spectral entropy distinguishes clean, tonal motion (energy in one band) from broadband, noise-like windows; the per-axis spectral-entropy features are among the strongest label correlates ($|r| \approx 0.47$–0.52 for the accelerometer), consistent with noise-like windows drifting more than smoothly manoeuvring ones.

2.2.3 Cross-axis and magnitude features (26)

A drone is a rigid body, so its axes are coupled; three groups of features capture coupling the per-axis statistics miss. Accelerometer and gyroscope *vector magnitudes* (mean, standard deviation, maximum of $\|a\|$ and $\|\omega\|$) give an orientation-independent measure of motion, invariant to the drone's heading. *Jerk* features (per-axis RMS and the mean and peak magnitude of $\mathrm{d}a/\mathrm{d}t$) target abrupt changes, which are precisely where discrete strapdown integration between samples is least accurate. Finally, the fifteen pairwise Pearson *correlations* between channels describe coordinated motion: strong cross-axis correlation is the signature of a banking turn or climb rather than uncorrelated sensor noise. On the 2 s windows the accelerometer pair `corr_ax_az` appears in the RandomForest ranking; accel–gyro cross terms are weaker than the accel means and RMS.

2.2.4 Scaling and the no-leakage constraint

Because the project splits by sequence (leave-one-sequence-out, Section 2.1), feature standardisation must respect fold boundaries: a `StandardScaler` is fit on each fold's training sequences only and then applied to the held-out sequence. Fitting a scaler across all data first would leak test-set statistics into training. The handoff tables `features_<seq>.csv` are therefore stored *unscaled*, and scaling is performed per fold at modelling time.

2.2.5 Redundancy and feature selection

The feature set is deliberately broad and contains known collinearities (variance is the square of standard deviation; RMS approximates the absolute mean when the mean dominates). Pooled across the two easy sequences, 22 feature pairs exceed a 0.95 absolute correlation. Tree-based models are insensitive to this redundancy, and a feature-selection experiment (notebook `04_feature_engineering`) confirms that a compact subset of the highest-ranked

features retains most of the predictive signal; the full set is retained for Phase 1 so that per-family contributions can be compared honestly in the ablation discussion.

3 Models

Every model maps a window's feature vector $\mathbf{x} \in \mathbb{R}^{104}$ (Section 2.2) to a predicted drift $\hat{y}$, the 2 s raw-IMU position error `err_mag_m` in metres. We compare a mean baseline with three linear models: ordinary least squares (OLS), Ridge and Lasso. Ridge and Lasso each have one penalty strength α, chosen on the training flight only (Section 3.3).

3.1 Baseline and linear models

Mean baseline. Predict the average training label for every window, $\hat{y} = \bar{y}_{\text{train}}$. It ignores the features, so a model that cannot beat it has learned nothing that transfers to a new flight. The label is about 1.2 m per window but varies by only 6–8 cm around that (Section 1), so the baseline is already close in absolute terms; R^2 measures how much of that 6–8 cm variation a model explains.

Linear regression (OLS). The linear model $\hat{y} = \mathbf{w}^\top \mathbf{x} + b$ that minimises the squared training error $\sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i - b)^2$, with nothing to tune. It is the unregularised reference: comparing it with Ridge and Lasso measures what their penalties buy.

Ridge regression. The same linear model, fitted by minimising

$$\sum_i \big(y_i - \mathbf{w}^\top \mathbf{x}_i - b\big)^2 + \alpha \|\mathbf{w}\|_2^2. \tag{6}$$

The L2 penalty shrinks every weight towards zero. It is needed because many features are near-duplicates (variance is the square of standard deviation; 22 feature pairs have $|r| > 0.95$ on the 2 s windows): least squares can give such a pair large weights of opposite sign that cancel on the training flight and fail on a new one. With 105 parameters and only 592–720 training windows, OLS has room to do exactly that. We search α over 17 values from 10^{-3} to 10^{5} (half-decade steps).

Lasso regression. The same model with an L1 penalty,

$$\frac{1}{2n} \sum_i \big(y_i - \mathbf{w}^\top \mathbf{x}_i - b\big)^2 + \alpha \|\mathbf{w}\|_1, \qquad \|\mathbf{w}\|_1 = \textstyle\sum_j |w_j|. \tag{7}$$

L2 shrinks every weight a little; L1 pushes weak weights to *exactly* zero, so Lasso also selects features, and from a group of near-duplicates it tends to keep one. Because the label's spread is only a few centimetres, the useful α values are small: at $\alpha = 0.1$ every weight is already zero and Lasso becomes the mean baseline. We search 16 values from 10^{-6} to 10^{-1}.

For OLS, Ridge and Lasso the features are standardised (zero mean, unit variance) inside the model pipeline, so the penalty treats every feature equally and the scaler's mean and standard deviation come from the training flight only.

3.2 Physics feature: the drift a calibrated bias explains

About 95% of the raw label is integrated IMU bias (Section 2.1). Both flights use the same sensor, so a bias measured on the *training* flight is a fair calibration constant for the test flight, as a real system reuses an earlier calibration. For each window we integrate the raw IMU twice

from the same ground-truth start state the label uses: once as-is, and once after subtracting the calibration bias $\bar{\mathbf{b}}$ (gyroscope and accelerometer, 6 values). The difference

$$\mathbf{d} = \hat{p}_{\text{end}}^{\text{raw}} - \hat{p}_{\text{end}}^{\bar{\mathbf{b}}} \tag{8}$$

is the position error that bias alone accounts for. Its components `phys_dx`, `phys_dy`, `phys_dz` and magnitude `phys_mag` form four extra features (`features/physics.py`). $\bar{\mathbf{b}}$ is the mean ground-truth bias of the fold's training flight, so each fold has its own table and the test flight's bias is never used. With the true per-window bias, $\mathbf{d}$ equals the raw minus bias-corrected label exactly (`tests/test_physics_features.py`). Like the label, the feature assumes the ground-truth start state of each window.

The feature is used without any fitting in Section 4.5 (predict `phys_mag` directly). It is *not* part of the 104 inputs of the models in Section 4.1; adding it is future work.

3.3 Evaluation protocol

Outer split: leave one flight out. Windows overlap by 87.5% (2 s length, 0.25 s stride), so a random train/test split would score a model on near-copies of its training windows. We split by flight instead (`splits.json`): train on MH_02_easy and test on MH_01_easy, then the reverse. Only these held-out scores are reported, and the test flight is used for nothing else.

Choosing α: time-block cross-validation. Each fold has a single training flight, so there is no second flight to validate on. We therefore cut the training flight into 5 contiguous time blocks. For each α, each block is scored by a model trained on the other four, and training windows within 16 windows (4 s, twice the window length) of the scored block are dropped, so no training window shares an IMU sample with a scored one (`tuning.py`). A random k-fold split would instead put overlapping neighbours on both sides and reward memorisation.

The median rule. The usual rule picks the α with the lowest *mean* error over the blocks. Here that fails: the first block of each flight is the take-off, where the drift is lower and its standard deviation 3–5× larger than in the rest of the flight (Figure 1). When that block is scored, the model has seen no take-off at all; for Lasso trained on MH_01 it misses by 9–17 cm, while the other blocks are at 2–5 cm. That one block dominates the mean, and the mean is smallest at large α, i.e. close to predicting the average. The final model is trained on the whole flight, take-off included, so that block's score is unrepresentative. We therefore pick the α with the lowest *median* block error, then refit on the whole training flight and score the test flight once. Section 4.2 checks this choice.

Metrics. On the held-out flight we report mean absolute error (MAE), root mean squared error (RMSE), bias $= \overline{\hat{y} - y}$, and $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$. $R^2 = 0$ means no better than knowing the *test* flight's mean; the mean baseline uses the *training* mean, so its R^2 is slightly negative. Training MAE is reported as well, to measure over-fitting.

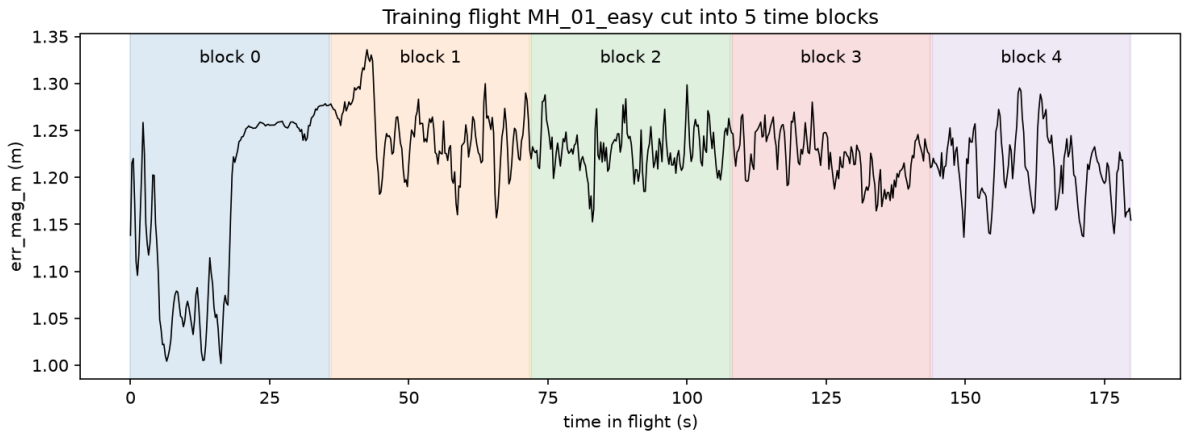


Figure 1: The training flight MH_01_easy cut into the 5 time blocks used to choose α. Block 0 contains the take-off, where the drift dips to about 1.0 m and varies far more than in blocks 1–4.

4 Results

All scores are on the held-out flight of each fold (Section 3.3); the target is `err_mag_m` and the inputs are the 104 Part C features.

4.1 Held-out performance

Table 2: Held-out MAE and R^2 for `err_mag_m`, by test flight. The model is trained on the other flight.

	MAE (cm)			R^2		
Model	MH_01	MH_02	Mean	MH_01	MH_02	Mean
Mean baseline	4.40	4.65	4.53	−0.04	−0.02	−0.03
OLS	3.12	2.87	2.99	0.40	0.76	0.58
Ridge (L2)	2.80	2.60	2.70	0.63	0.80	0.72
Lasso (L1)	2.59	2.42	2.50	0.68	0.84	0.76

Table 2 gives the headline result. Every model beats the mean baseline on both flights. Lasso is best in both folds: its mean MAE of 2.50 cm is 45% below the baseline's 4.53 cm, and it explains 76% of the test flight's drift variation on average. Ridge is close behind (2.70 cm, $R^2 = 0.72$); OLS gains least (2.99 cm, $R^2 = 0.58$) and is weakest on MH_01 ($R^2 = 0.40$), where it is trained on the shorter flight. RMSE gives the same order (fold means: baseline 6.91, OLS 4.15, Ridge 3.49, Lasso 3.23 cm), and all models are nearly unbiased: the largest mean bias is −0.49 cm (OLS), and Lasso's is −0.12 cm. The baseline's bias is ±1.17 cm per fold because the two flights' mean drifts differ by that much.

4.2 Choosing α

The median rule chose $\alpha = 316$ (trained on MH_02) and 31.6 (trained on MH_01) for Ridge, and $\alpha = 0.0022$ and 0.00046 for Lasso.

To check these choices we swept α over the whole grid and scored the *test* flights (Figure 3). This uses test data, so it is a diagnostic only and played no part in the choice. The curves show the expected trade-off: at small α the model is essentially OLS, with low training error but high held-out error; at large α every weight is pushed to zero and the model becomes the mean

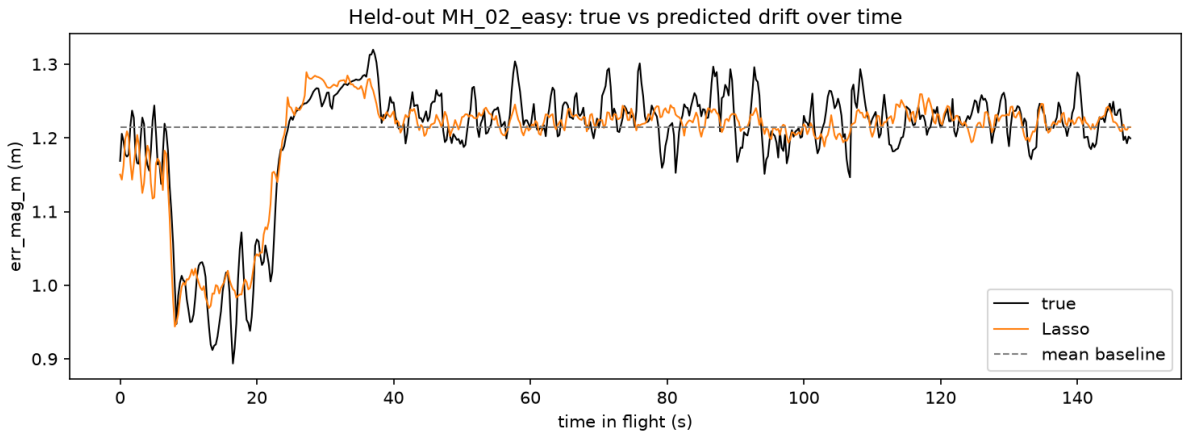


Figure 2: True drift and Lasso's prediction over the held-out MH_02_easy flight (trained on MH_01_easy). Lasso follows the large take-off dip in the first ∼25 s, but in normal flight its prediction is much smoother than the truth.

baseline. The held-out minimum lies at $\alpha \approx 100$–300 for Ridge and $\alpha \approx 0.002$–0.005 for Lasso on both flights. All four chosen values fall in the low part of these curves.

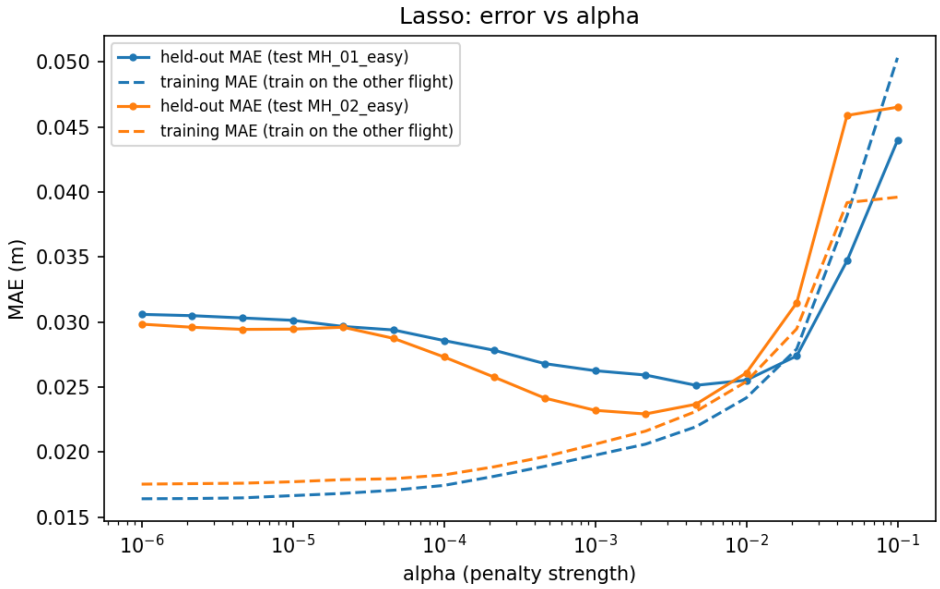


Figure 3: Lasso training and held-out MAE against α, for both folds (diagnostic only; α is not chosen from this). Ridge shows the same shape (`figures/ml/alpha_sweep_ridge.png`).

The median rule matters. With the same 5 blocks, the usual mean rule, trained on MH_01, chose Ridge $\alpha = 3.2 \times 10^4$ (held-out $R^2 = 0.32$ instead of 0.80). The one-standard-error rule was worse: it chose the largest α in the grid for both Ridge ($R^2 = 0.13$) and Lasso, where $\alpha = 0.1$ zeroes every weight ($R^2 = -0.02$). Trained on MH_02, all three rules gave good choices. With 3, 5, 8 or 10 blocks, the median rule never chose an α that zeroes every Lasso weight; with 5 or more blocks it gave Lasso $R^2 = 0.68$ on MH_01 and 0.84–0.85 on MH_02. Tuning on a single flight remains noisy, though: the median curve itself is bumpy (notebook 05).

Table 3: Training vs. held-out MAE, mean of the two folds.

Model	Train MAE (cm)	Held-out MAE (cm)	Held-out / train
Mean baseline	4.50	4.53	1.01
OLS	1.65	2.99	1.81
Ridge (L2)	1.95	2.70	1.38
Lasso (L1)	2.01	2.50	1.24

4.3 Over-fitting

OLS fits its training flight almost twice as well as the new flight (Table 3): with 104 correlated features and about 600 windows it learns flight-specific noise. The penalties trade a slightly worse training fit for a clearly better held-out one: the held-out/train ratio falls from 1.81 to 1.38 (Ridge) and 1.24 (Lasso). Lasso does this with few features: it keeps 18 of the 104 when trained on MH_02 and 40 when trained on MH_01, and 9 are kept in both folds, including the accelerometer means and RMS, which encode the window's tilt against gravity (notebook 05).

4.4 What the models capture

Table 4: Lasso's held-out R^2 and MAE in the first 20% of each flight (take-off) and in the remaining 80%, with the mean baseline's MAE.

Test flight	Part	Lasso R^2	Lasso MAE (cm)	Baseline MAE (cm)
MH_01	first 20%	0.86	2.85	8.77
	remaining 80%	0.10	2.53	3.30
MH_02	first 20%	0.84	3.60	13.03
	remaining 80%	0.17	2.12	2.57

The R^2 in Table 2 has to be read with care. Most of a flight's drift variation is the take-off dip, from about 1.22 m down to about 1.0 m (Figure 2). The models track this slow, large change well: Lasso reaches $R^2 = 0.84$–0.86 in the first 20% of each flight (Table 4), and that segment dominates the whole-flight R^2. In normal flight Lasso explains only a small part of the fast window-to-window changes ($R^2 = 0.10$–0.17); its predictions are much smoother than the truth. It still beats the baseline there in MAE (2.53 vs 3.30 cm on MH_01, 2.12 vs 2.57 cm on MH_02). Ridge and OLS show the same pattern (R^2 0.50–0.77 in the first 20%, -0.17 to 0.11 in the rest).

4.5 Window length

Table 5: Mean held-out R^2 of `err_mag_m` over the two folds, by window length, for bias-based predictors.

Window length	0.5 s	1 s	1.5 s	2 s
Physics feature, no fitting	0.06	0.47	0.70	0.82
Physics feature, linear fit	0.13	0.48	0.70	0.82
Oracle ceiling (true bias)	0.15	0.51	0.72	0.83

The first version of this pipeline used 1 s windows. Because the raw label is mostly integrated bias, whose position error grows quickly with time, while the rest behaves like noise from window to window, a longer window should make the label more predictable. To test this without rebuilding the 104 features, every flight was re-windowed at 0.5–2 s and scored with three bias-based predictors (Table 5): the physics feature `phys_mag` used directly (Section 3.2); a linear fit $y \approx a + c \cdot$ `phys_mag` on the training flight; and an *oracle* fit on the test flight's own drift from

its true bias, which no real model can know and which bounds what a bias-based predictor can reach. All three rise steadily with window length; at 1 s even the oracle explains only about half of the variation, against 0.83 at 2 s. This is why the window was changed to 2 s.

The table also shows that the physics feature alone, with no fitting, reaches $R^2 = 0.82$ at 2 s, above Lasso's 0.76 on the 104 motion features (Table 2). Adding it to the models is the obvious next step. Like the label, it uses the ground-truth state at the start of each window.

4.6 Limitations

With only two flights, every result rests on two folds, and each model is tuned on a single flight. Both flights are "easy" Machine Hall sequences with the same sensor, so the results say nothing yet about faster or more aggressive flights. And as Section 4.4 shows, much of the explained variation comes from the take-off.

References

[1] Michael Burri, Janosch Nikolic, Pascal Gohl, Thomas Schneider, Joern Rehder, Sammy Omari, Markus W. Achtelik, and Roland Siegwart. The EuRoC micro aerial vehicle datasets. *The International Journal of Robotics Research*, 35(10):1157–1163, 2016.